

Ryan Bowering | Curriculum Vitae

rnbowering@gmail.com | (847) 417-3797 | bowerim.github.io | GitHub [@bowerim](#) | LinkedIn [@rnbowering](#)

Research Interests

- Active acoustic control with flat-panel loudspeakers
- Generative modeling
- Music information retrieval

Education

University of Rochester (Rochester, NY) Aug. 2025 - Present
PhD in Computer Science, AI Concentration GPA: 4.0

Rose-Hulman Institute of Technology (Terre Haute, IN) Sept. 2021 - May 2024
BS in Computer Science, Minors in AI, Math, Data Science, Music GPA: 3.91

Professional Experience

Los Alamos National Laboratory (Los Alamos, NM) July 2024 - July 2025
Post-Bac Researcher, Data Analysis

- Developed generative models for shock waveform synthesis, achieving improved spectral fidelity and orders-of-magnitude faster inference than classical methods
- Co-authored paper on inverse reconstruction of shock time series ([arXiv:2603.03229](#))
- Built GPU-accelerated pipeline for large-scale synthetic data generation

Cummins Inc. (Columbus, IN) June 2023 - Aug. 2023
Software Engineering Intern

- Completely automated the External Recipient registration process for update notifications
- Applied Scrum/Kanban and used Python, React.js and AWS Lambda and APIGateway

Publications

[1] A. Watts, A. Jeon, D. Newton, and **R. Bowering**, "Inverse reconstruction of shock time series from shock response spectrum curves using machine learning," [arXiv:2603.03229](#), 2026.

Teaching

Lead TA, CSC161 (Intro to Computer Science) Spring 2026
University of Rochester

- Managed team of 5 other lab TAs to run labs and grade assignments
- Helped proctor exams

TA/Grader, MA474 (Theory of Computation) Winter 2023
Rose-Hulman Institute of Technology

- Graded assignments for the professor
- Held in-person office hours to help students with classwork

Projects

Alert Infrastructure for IOT Devices (CSSE Department Award) *Sept. 2023 - May 2024*

- Created an infrastructure-as-code system in Python to streamline writing alerts for IOT devices
- Built a Slack bot to send alert notifications and interact with data in Snowflake

AI-Generated Image Detector [GitHub](#) *April 2023 - Nov 2023*

- Trained neural networks with Python to classify real and AI-generated images
- Achieved a testing accuracy of 98% by fine-tuning a Vision Transformer model

Evolutionary Data Modeler [GitHub](#) *Aug. 2023 - Sept. 2023*

- Evolves equations to model input data with a tunable genetic algorithm
- Derived Kepler's third law using his available planetary data

Technical Skills

- Languages: Python, C++, C, Java, SQL, MATLAB, JavaScript
- Frameworks: PyTorch, Pandas, Scikit-Learn

Activities

- *University of Rochester Rowing Team* *2025 - Present*
- *University of Rochester Swim Team* *2025 - 2026*